LSci 51/CogS 56L: Acquisition of Language

Lecture 15

Syntactic acquisition II

Development of comprehension

Clever non-syntactic strategies

Clever comprehension strategies children use:

Recognized words in utterance: “Hoggle”, “Jareth”, “scared”

Use **world knowledge** to figure out likely sequence of events (Blything et al. 2015, Wagner & Holt 2023).

Jareth

Hoggle

Works really well for sentences that are predictable from world knowledge (in a world where Jareth is often doing the scaring and Hoggle is often being scared):

“Jareth scared Hoggle.”

…but not so well for ones where the events are not predictable from world knowledge:

“Hoggle scared Jareth.”

Clever non-syntactic strategies

Clever comprehension strategies children use:

Recognized words in utterance: “cowered”, “threw”, “disguise”, “Hoggle”, “Jareth”

Use the **order** of words to predict who did what to whom (Clark 1971, DeRuiter et al. 2018, Wagner & Holt 2023): first mention is what happened first.

Works really well for sentences where order-of-mention is the order of action:

“Jareth threw off his disguise before Hoggle cowered.”

Actual event: Jareth-throw-disguise, then Hoggle-cower. [Matches word order.]

Jareth

Hoggle

Clever non-syntactic strategies

Clever comprehension strategies children use:

Recognized words in utterance: “cowered”, “threw”, “disguise”, “Hoggle”, “Jareth”

Use the **order** of words to predict who did what to whom (Clark 1971, DeRuiter et al. 2018, Wagner & Holt 2023): first mention is what happened first.

…but not so well for ones where it’s not:

“Hoggle cowered after Jareth threw off his disguise.”

Actual event: Jareth-throw-disguise, then Hoggle-cower. [Does not match word order]

Jareth

Hoggle

Clever non-syntactic strategies

Clever comprehension strategies children use:

Recognized words in utterance: “knight”, “dwarf”, “bump”

Use the **order** of words to predict who did what to whom: first mention is the one doing the action, second mention is the one the action is done to.

Works really well for active sentences:

“The **knight bump**ed the **dwarf**.”

Actual event: knight-bumps-dwarf

[Matches word order]

Clever non-syntactic strategies

Clever comprehension strategies children use:

Recognized words in utterance: “knight”, “dwarf”, “bump”

Use the **order** of words to predict who did what to whom: first mention is the one doing the action, second mention is the one the action is done to.

…but not so well for passives:

“The **knight** was **bump**ed by the **dwarf**.”

Actual event: dwarf-bumps-knight

[Does not match word order]

knight

dwarf

Passives

http://arnoldzwicky.org/2012/02/22/misfired-indirecion/

[Extra]

Passives

http://www.thelingspace.com/episode-39
https://www.youtube.com/watch?v=NJ5lLNBabGc
1:59 - 4:45

Passives

Passives are tricky because the subject of the sentence is the "done-to" of the action (rather than the "doer" as it is in active sentences).

Active sentence:

Sarah saved Toby.

Subject Verb Object

semantically "light" verb

Passive equivalent:

Toby was saved by Sarah.

Subject Verb

doer "by" phrase

Passives

English-speaking children usually start producing passives when around age 3.

Some example passives & the ages when they were produced:

“Do you think the flower’s supposed to be picked by somebody?” (2;10)

“So it can’t be cleaned?” (3;2)

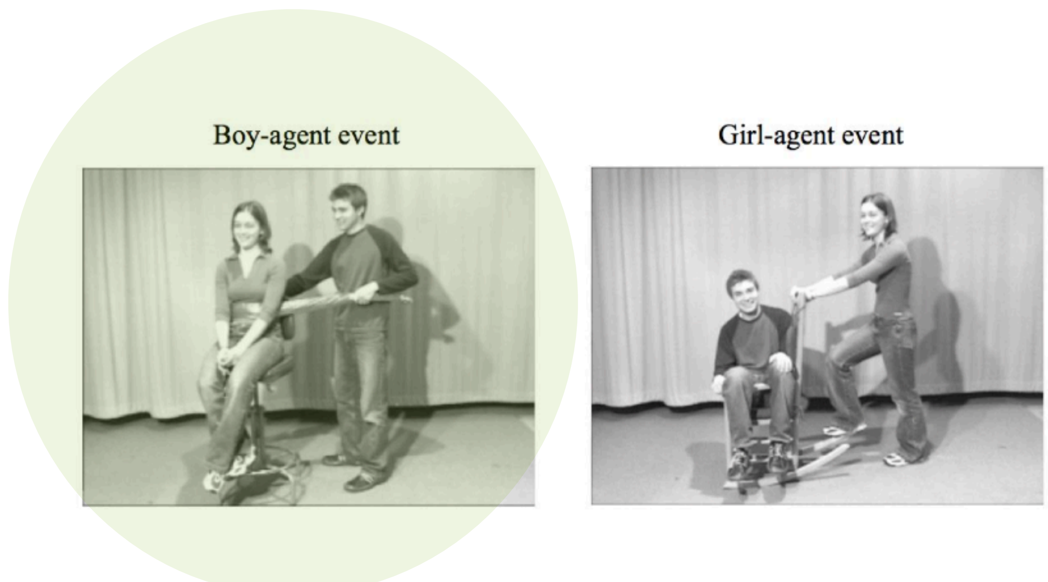
“I don’t want the bird to get eated.” (3;7)

“She brought her inside so she won’t get all stinked up by the skunk.” (4;1)

Passives

English-speaking three-year-olds also seem to comprehend novel passives when they're given enough time to process the sentences (Messenger & Fisher 2018). This suggests English-speaking three-year-olds do understand the structure.

"The girl is getting snedded by the boy!"



Passives

In fact, children seem to over-produce passives, applying a “passive” rule to verbs that (some) adults wouldn’t make passive.

Passive rule =~ be/get + VERB + en/ed

Some example “over-produced” passives:

“…they won’t get staled.” (3;6)

“The tiger will come and eat David and then he will be died.” (4;0)

“Why is the laundry place stayed open all night?” (4;3)

Passives

Still, despite producing passives spontaneously, children seem to have persistent trouble understanding some passive sentences.

Standard comprehension task with
reversible passive:

Hoggle was hugged by Sarah.

reversible passive:
Makes sense either way, so kids can't rely on world knowledge

Hoggle-hug-Sarah or Sarah-hug-Hoggle
✓ ✓

Passives

Still, despite producing passives spontaneously, children seem to have persistent trouble understanding some passive sentences.

Standard comprehension task with
reversible passive:

4 years old

✔ *Hoggle was pushed by Sarah.*

✗ *Hoggle was remembered by Sarah.*

Passives

Still, despite producing passives spontaneously, children seem to have persistent trouble understanding some passive sentences.

Nguyen & Pearl 2018, 2019, 2021, Liter & Lidz 2021, Agostino, Gavarró, & Sanots 2024: This seems like it has to do with the semantic features of the verbs. Certain features are more salient at different ages.

carry, push, kiss

4 years old

+actional

✔ *Hoggle was pushed by Sarah.*

✗ *Hoggle was remembered by Sarah.*

+mental state

where the subject experiences the mental state in an active sentence and nothing really "happens" to the object

Sarah VERBed Hoggle.

forget, know, believe

Passives

Eventually, children learn to notice the more subtle signals of a passive sentence – the light verb, the participle (-en/-ed) ending, and sometimes the “doer” by phrase.

“Hoggle was remembered by Sarah”

But it *does* take awhile…(sometimes up till 9 years old!)

Passives

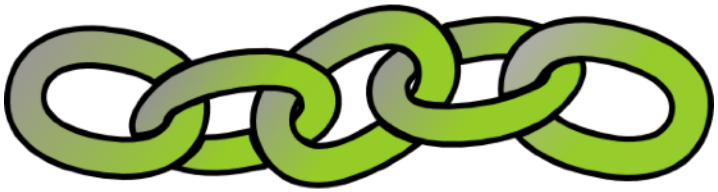
Eventually, children learn to notice the more subtle signals of a passive sentence – the light verb, the participle (-en/-ed) ending, and sometimes the “doer” by phrase.

“Hoggle was remembered by Sarah”

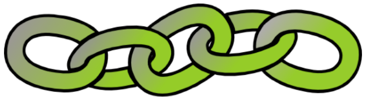
Interestingly, passive usage doesn’t differ much in frequency between child-directed speech and children’s books (Altmiller, Corriveau, & Arunachalam 2022). So, just being exposed to “more sophisticated” book language, the way older children are, isn’t likely to help a lot.

A more general acquisition problem: Learning how verbs behave

The Linking Problem:
Children must learn how verbs behave — that is, how verbs and verb arguments of their language link up to create meaning.

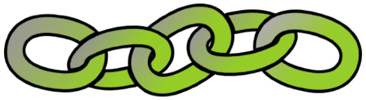


The Linking Problem

What is this likely to mean? 

The little girl *blicked* the kitten on the stairs.

The Linking Problem

What is this likely to mean? 

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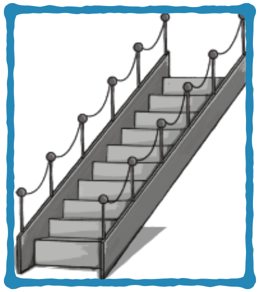


Event participants

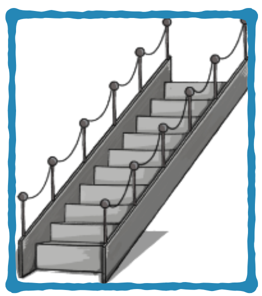
The Linking Problem

What is this likely to mean?

Syntactic positions

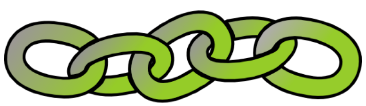
Subject | Object | Oblique Object

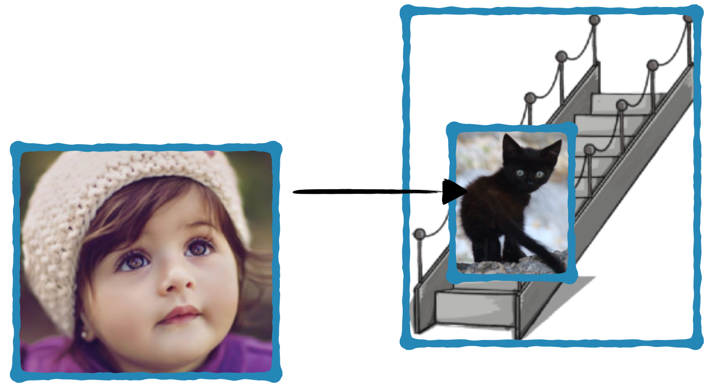
The little girl *blicked* the kitten on the stairs.



Event participants

The Linking Problem





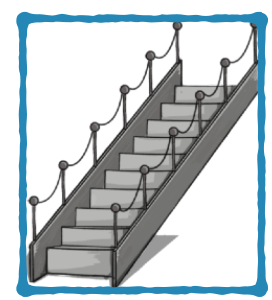
Syntactic positions

Subject

Object

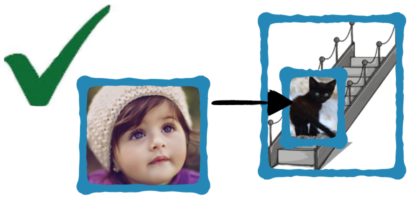
Oblique Object

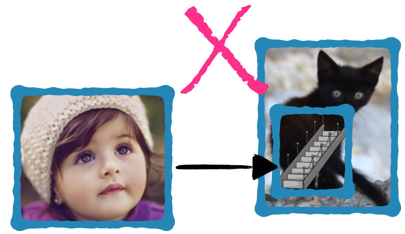
The little girl *blicked* the kitten on the stairs.

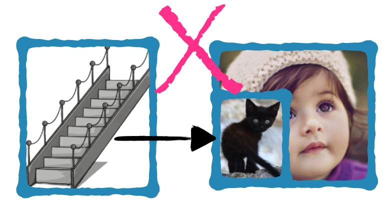


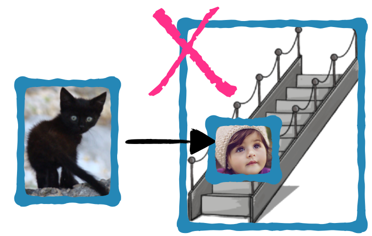
Event participants

The Linking Problem

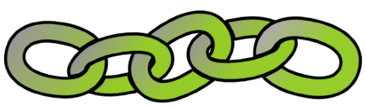








…compared to these.



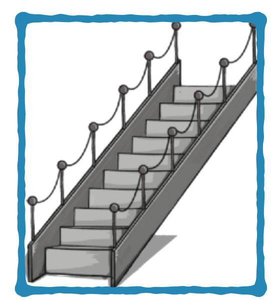
Syntactic positions

Subject

Object

Oblique Object

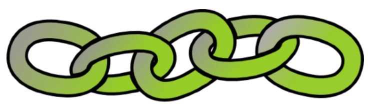
The little girl *blicked* the kitten on the stairs.

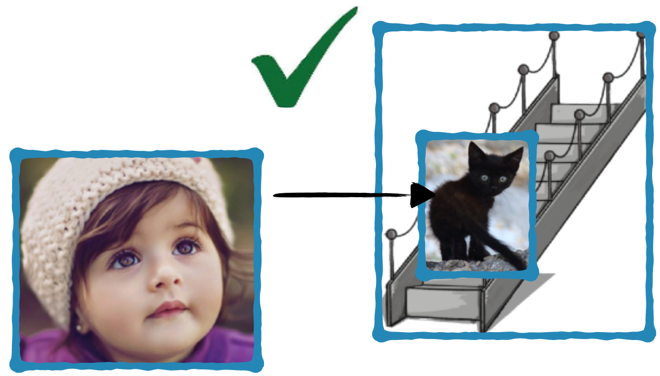


Event participants

The Linking Problem

Why?





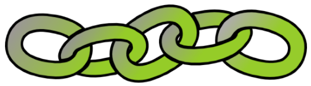
Syntactic positions

Subject | Object | Oblique Object

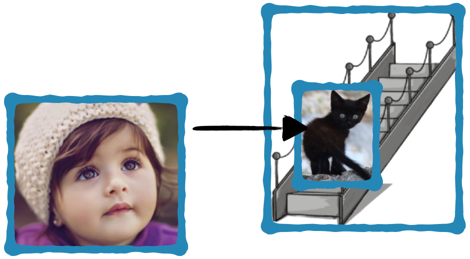
The little girl *blicked* the kitten on the stairs.

Event participants

The Linking Problem solved



We as adults have linking theories that help us interpret verbs in combination with their arguments.



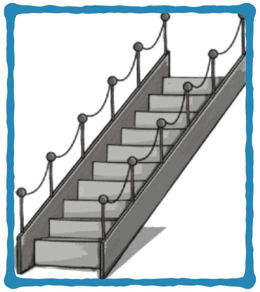
Syntactic positions

Subject

Object

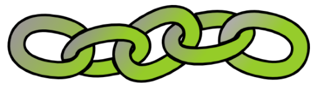
Oblique Object

The little girl *blicked* the kitten on the stairs.

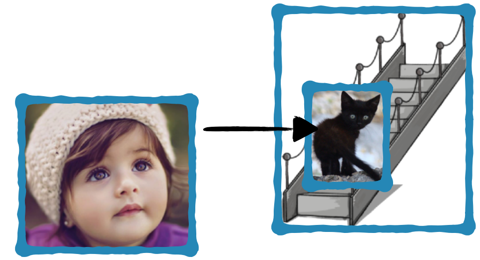


Event participants

The Linking Problem solved



We can also use these linking theories to produce verbs in combination with their arguments when we want to express a particular meaning.



Syntactic positions

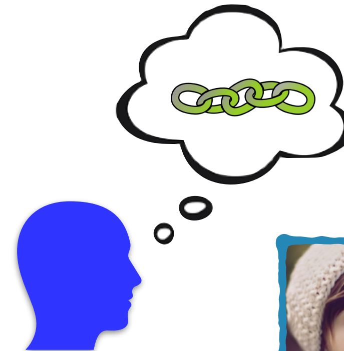
Subject | Object | Oblique Object

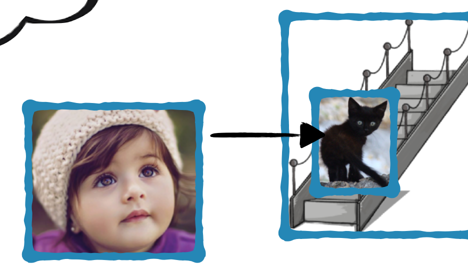
The little girl *blicked* the kitten on the stairs.

Event participants

The Linking Problem solved

These linking theories are mental representations that we as adults have developed. They let us link event participants and syntactic positions, so we know how to interpret an utterance — even when we don't know what the verb means.

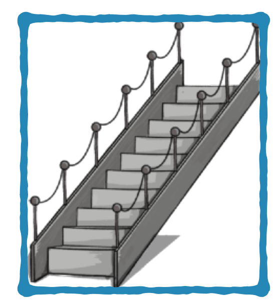




Syntactic positions

Subject | Object | Oblique Object

The little girl *blicked* the kitten on the stairs.



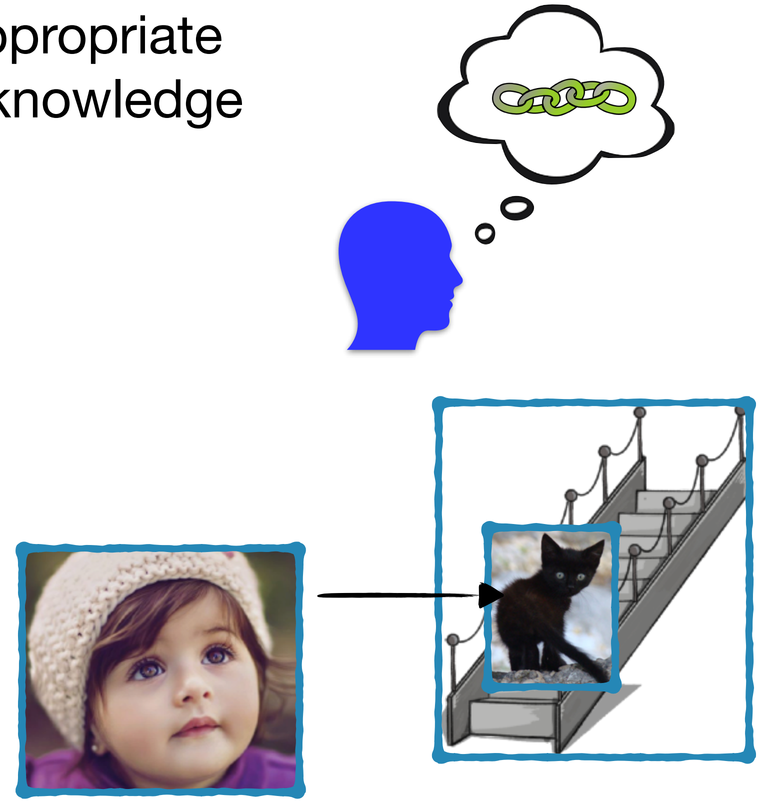
Event participants

Verbs, their arguments,
& linking meaning to structure

https://www.youtube.com/watch?v=thFkoo1YmW0
3:30-6:15

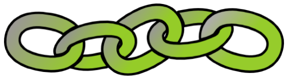
The Linking Problem solved

Children have to develop the appropriate linking theories as part of their knowledge of their native language.



The little girl *blicked* the kitten on the stairs.

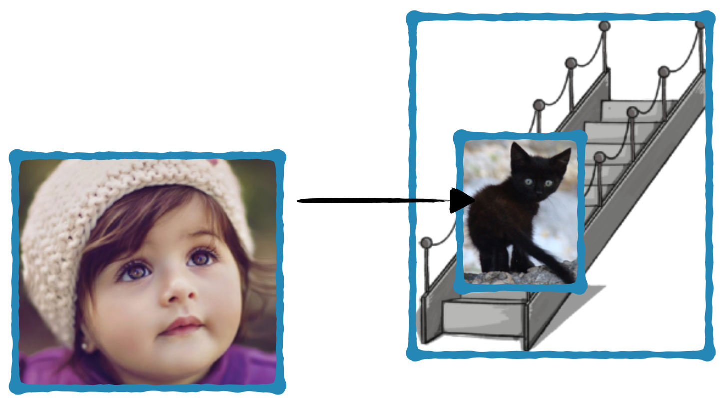
Linking theories



What does a linking theory look like?

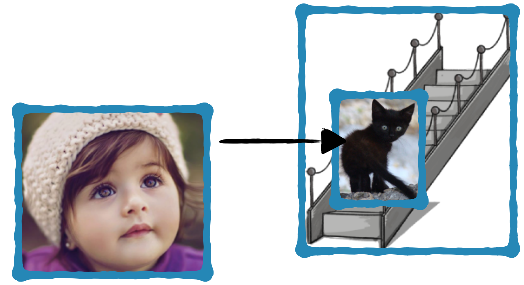
Subject Object Oblique Object

The little girl *blicked* the kitten on the stairs.



Linking theories

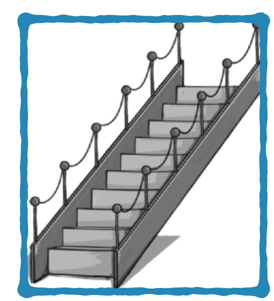
A linking theory links syntactic positions and thematic roles, which are participants in the event described.



The little girl *blicked* the kitten on the stairs.

Syntactic positions | Subject | Object | Oblique Object

Event participant roles
=
Thematic roles



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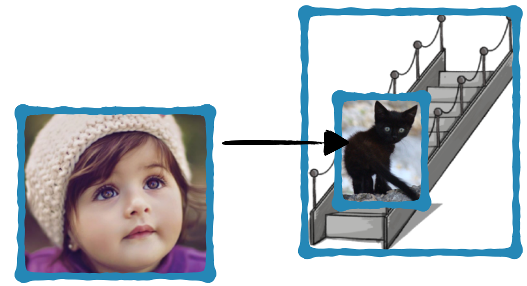


The little girl *blicked* the kitten on the stairs.

Syntactic positions

Subject

Object

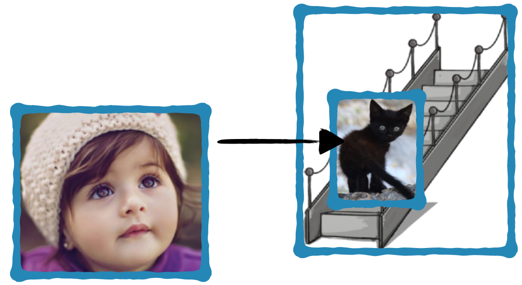
Oblique Object

Event participant roles
=
Thematic roles

Agent, Experiencer, Patient, Theme, Goal, Source, Location…

Linking theories

Current linking theory proposals involve mapping to an intermediate representation before mapping to the syntactic positions.



The little girl *blicked* the kitten on the stairs.

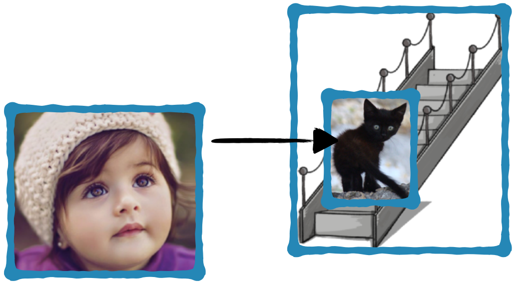
Syntactic positions Subject Object Oblique Object

Intermediate representations

Event participant roles
=
Thematic roles

Agent, Experiencer, Patient, Theme, Goal, Source, Location…

Some linking theory proposals



The little girl *blicked* the kitten on the stairs.

Syntactic positions — Subject — Object — Oblique Object

Intermediate representations

One idea: UTAH

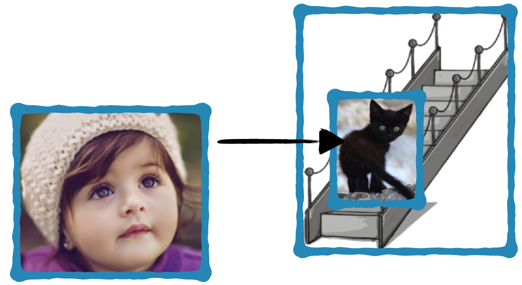
The **U**niformity of **T**heta **A**ssignment **H**ypothesis

Baker 1988, Baker 1997, Dowty 1991, Fillmore 1968, Grimshaw 1990, Jackendoff 1987, Perlmutter & Postal 1984, Speas 1990

Event participant roles
=
Thematic roles

Agent, Experiencer, Patient, Theme, Goal, Source, Location…

Some linking theory proposals



The little girl *blicked* the kitten on the stairs.

Syntactic positions — Subject — Object — Oblique Object

Intermediate representations

UTAH

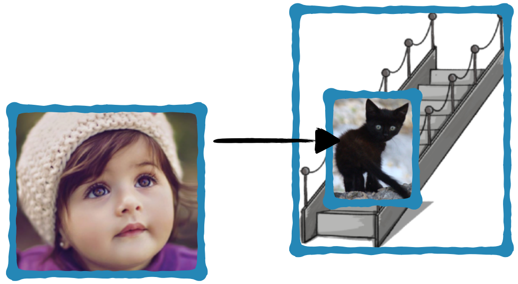
Thematic roles map to one of three **fixed macro-roles.**

proto-Agent	proto-Patient	Other

Event participant roles
=
Thematic roles

Agent, Experiencer, Patient, Theme, Goal, Source, Location…

Some linking theory proposals



The little girl *blicked* the kitten on the stairs.

Syntactic positions

Subject

Object

Oblique Object

These map to syntactic positions.

UTAH

fixed

Intermediate representations

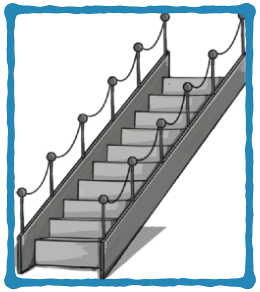
Event participant roles
=
Thematic roles

Agent, Experiencer, Patient, Theme, Goal, Source, Location…

Some linking theory proposals

UTAH in action

The little girl *blicked* the kitten on the stairs.



Some linking theory proposals

UTAH in action

The little girl *blicked* the kitten on the stairs.

Subject Object Oblique Object

Some linking theory proposals

UTAH in action

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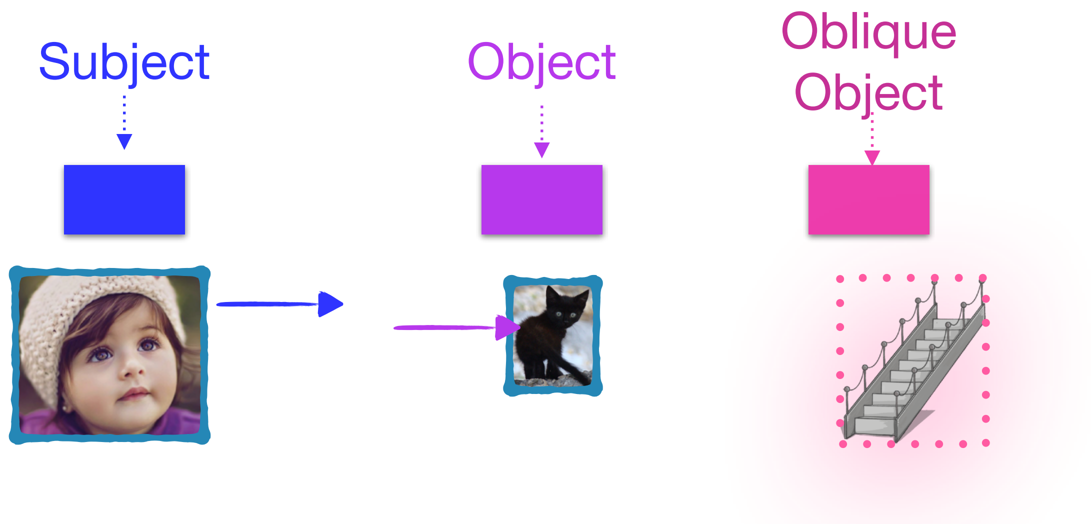

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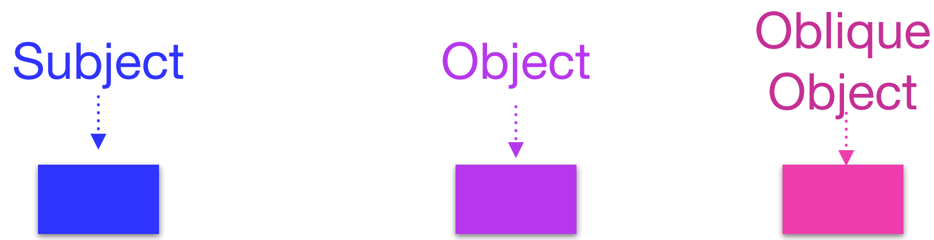
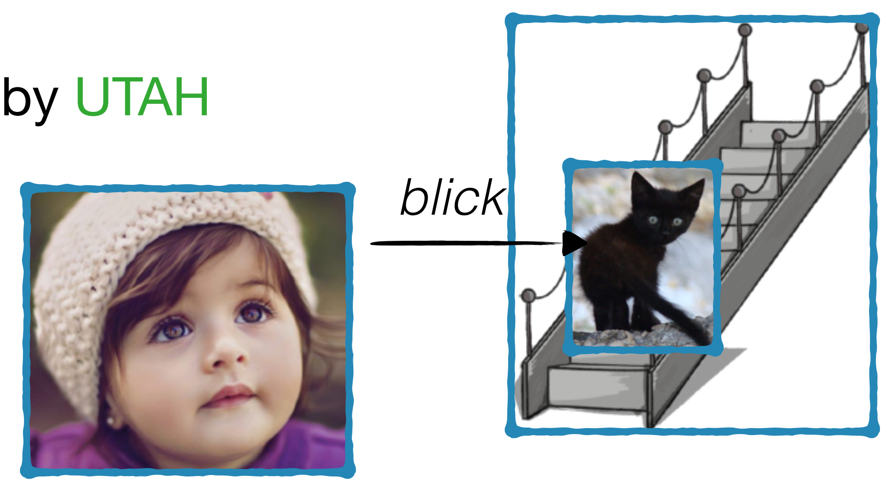


Predicted by UTAH



Some linking theory proposals

UTAH in action

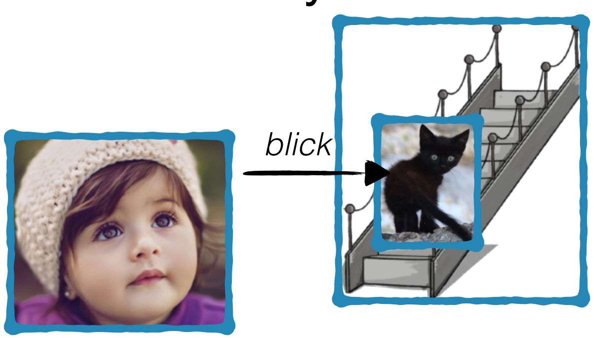
The little girl *blicked* the kitten on the stairs.

Subject

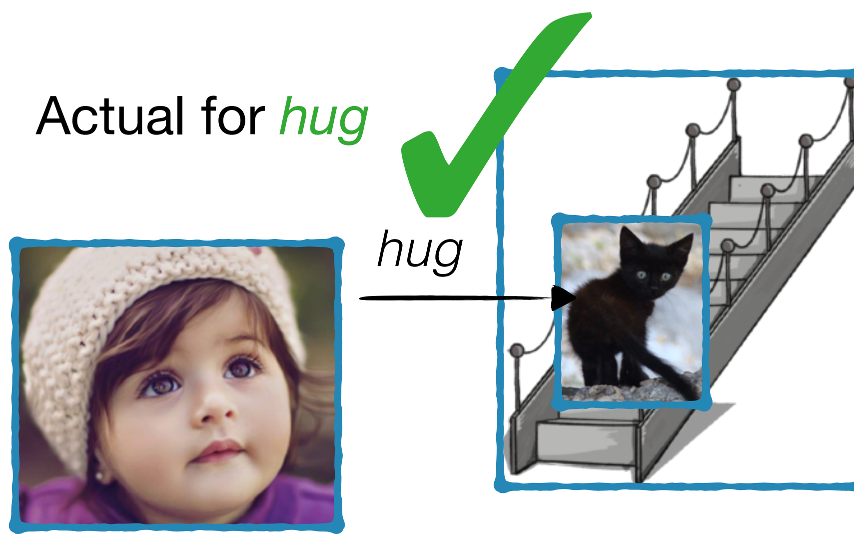
Object

Oblique
Object

Predicted by UTAH



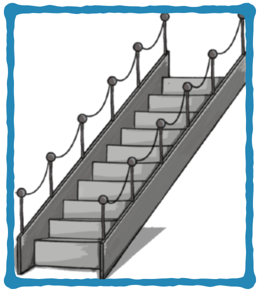
Actual for *hug*



Some linking theory proposals

UTAH in action
though sometimes there are exceptions, like the English passive.
Children would have to learn these exceptions, which is maybe why the passive takes them awhile to figure out.

The kitten was *blicked* on the stairs.



Some linking theory proposals

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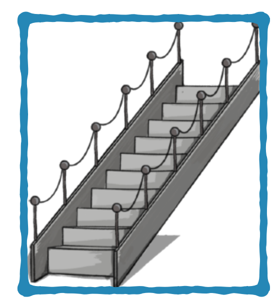
Oblique Object



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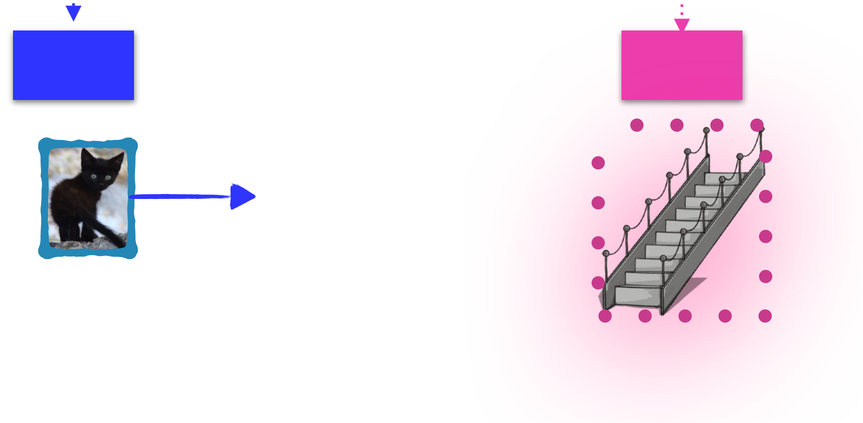

Some linking theory proposals

UTAH in action

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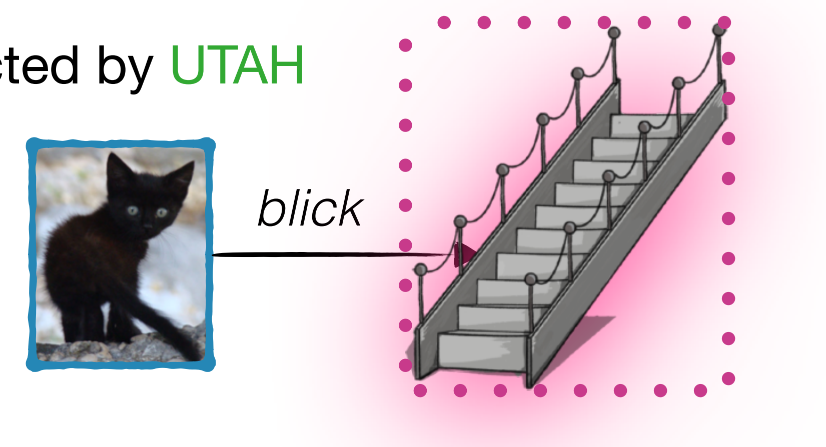
Subject

Oblique Object

Predicted by UTAH



Some linking theory proposals

UTAH in action

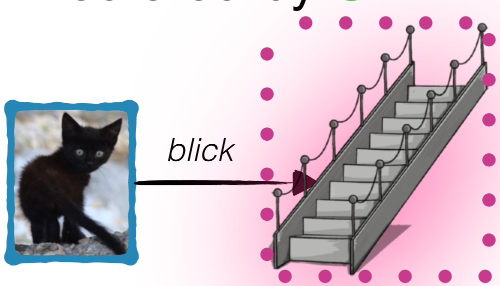
though sometimes there are exceptions, like the English passive. Children would have to learn these exceptions, which is maybe why the passive takes them awhile to figure out.

The kitten was *blicked* on the stairs.

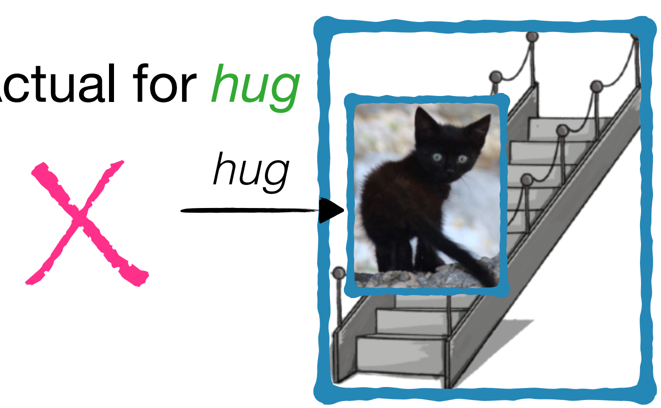
Subject

Oblique Object

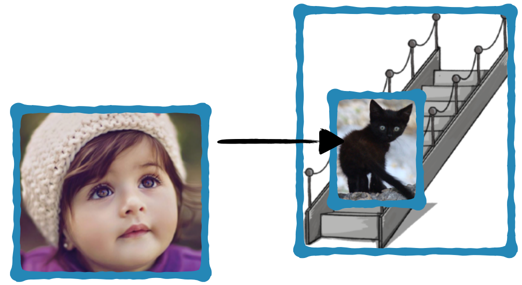
Predicted by UTAH



Actual for *hug*



Some linking theory proposals



The little girl *blicked* the kitten on the stairs.

Syntactic positions

Intermediate representations

Event participant roles
=
Thematic roles

Subject

Object

Oblique Object

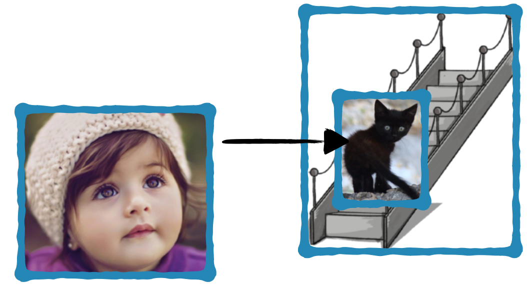
These map to syntactic positions.

UTAH

fixed

Agent, Experiencer, Patient, Theme, Goal, Source, Location…

Some linking theory proposals



The little girl *blicked* the kitten on the stairs.

Syntactic positions

Subject

Object

Oblique Object

Intermediate representations

UTAH

fixed

Another proposal

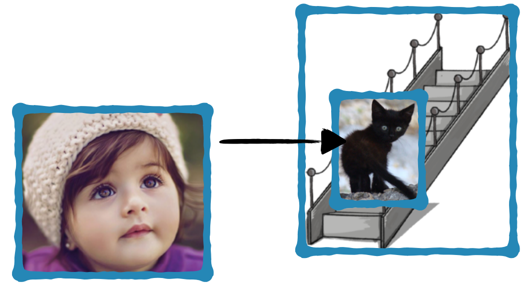
The (**r**elativized) **UTAH**

Larson 1988, Larson 1990

Event participant roles
=
Thematic roles

Agent, Experiencer, Patient, Theme, Goal, Source, Location…

Some linking theory proposals



The little girl *blicked* the kitten on the stairs.

Syntactic positions

Subject

Object

Oblique Object

Intermediate representations

UTAH

fixed

rUTAH

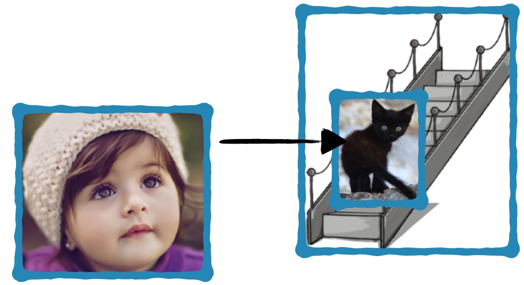
Thematic roles are **ordered relative** to each other.

Agent > Experiencer >
Theme > Patient >
(Source, Goal, Location)

Event participant roles
=
Thematic roles

Agent, Experiencer, Patient, Theme, Goal, Source, Location…

Some linking theory proposals



The little girl *blicked* the kitten on the stairs.

Syntactic positions

Subject

Object

Oblique Object

Intermediate representations

Whichever ones are present **map in order** to the available syntactic positions.

UTAH

fixed

rUTAH

relative

Agent > Experiencer >
Theme > Patient >
(Source, Goal, Location)

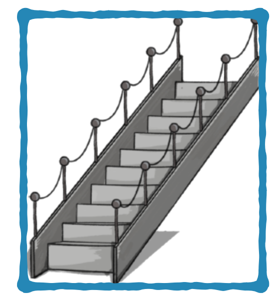
Event participant roles
=
Thematic roles

Agent, Experiencer, Patient, Theme, Goal, Source, Location…

Some linking theory proposals

rUTAH in action

The little girl *blicked* the kitten on the stairs.

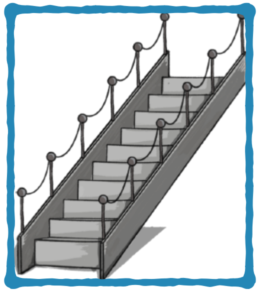


Some linking theory proposals

rUTAH in action

The little girl *blicked* the kitten on the stairs.

1st 2nd 3rd



Some linking theory proposals

rUTAH in action

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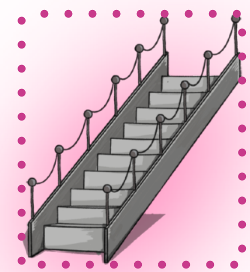
1st 2nd 3rd

Agent > Experiencer >
Theme > Patient >
(Source, Goal, Location)



Some linking theory proposals

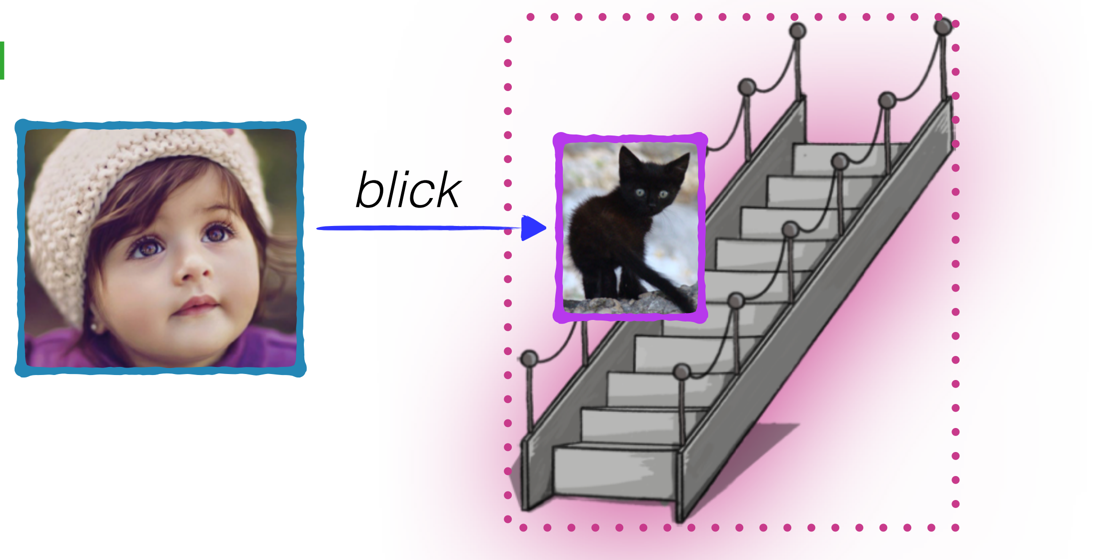
rUTAH in action

The little girl *blicked* the kitten on the stairs.

1st 2nd 3rd

Agent > Experiencer >
Theme > Patient >
(Source, Goal, Location)

one prediction by rUTAH



Some linking theory proposals

rUTAH in action

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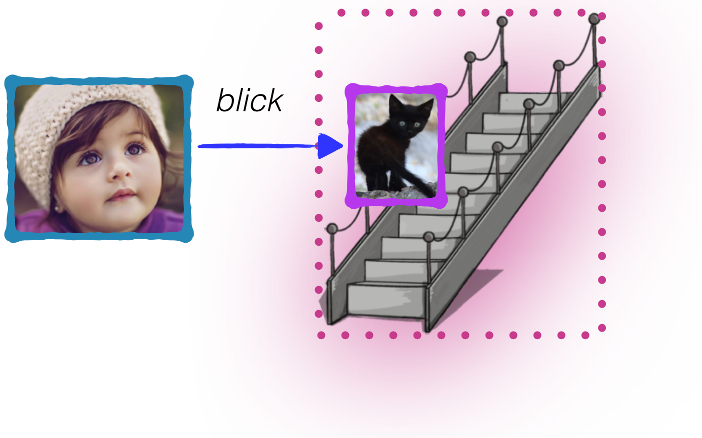

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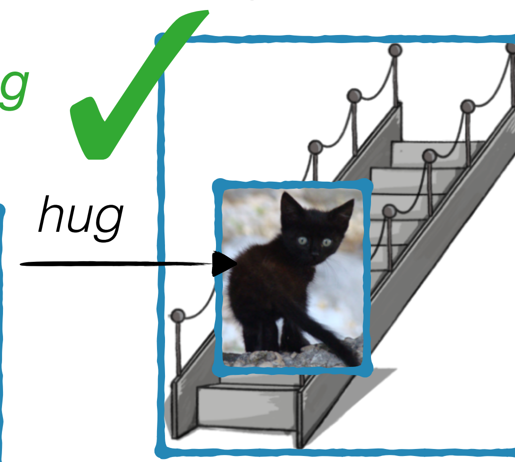
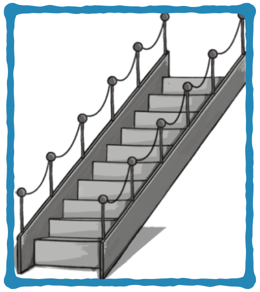


Some linking theory proposals

rUTAH in action

has fewer exceptions because of its flexibility. For instance, it can handle some forms of the English passive.

The kitten was *blicked* on the stairs.



Some linking theory proposals

rUTAH in action

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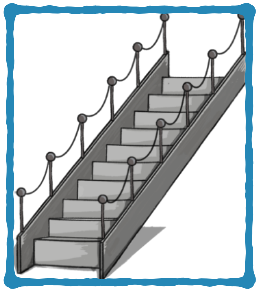
The kitten was *blicked* on the stairs.

1st

2nd



Some linking theory proposals

rUTAH in action

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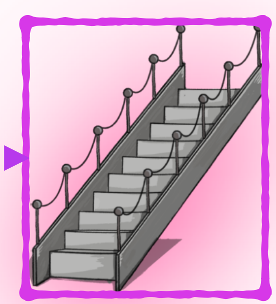
The kitten was *blicked* on the stairs.

1st

2nd

Agent > Experiencer >
Theme > Patient >
(Source, Goal, Location)



Some linking theory proposals

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The kitten was *blicked* on the stairs.

1st

2nd

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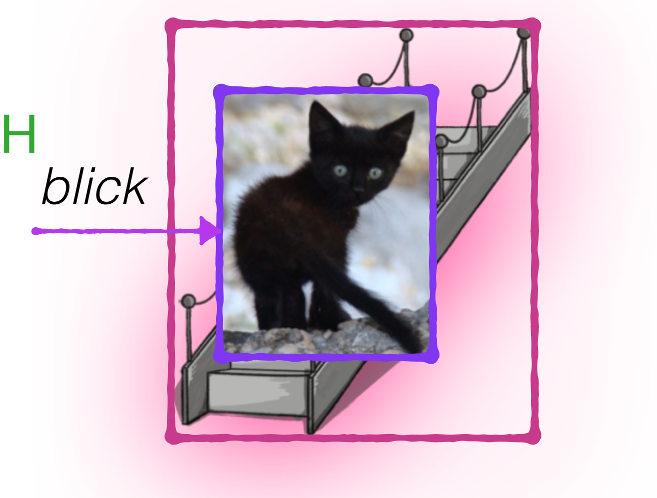
Theme > Patient >
(Source, Goal, Location)

one prediction by rUTAH



Some linking theory proposals

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2nd

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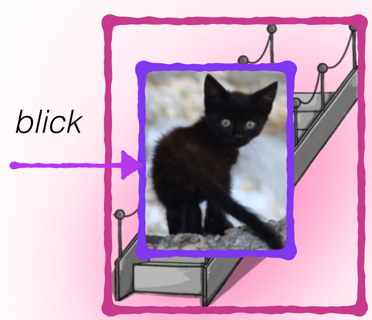
one prediction by rUTAH



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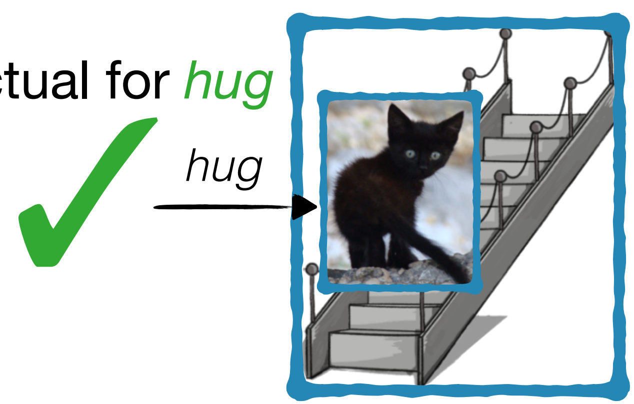
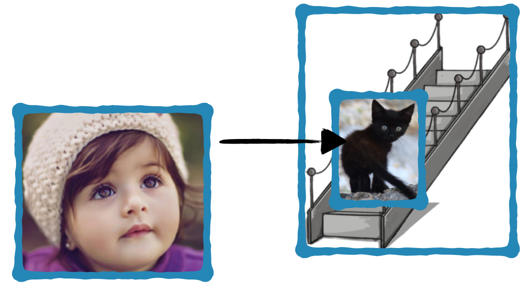


Some linking theory proposals: Recap



The little girl *blicked* the kitten on the stairs.

Syntactic positions

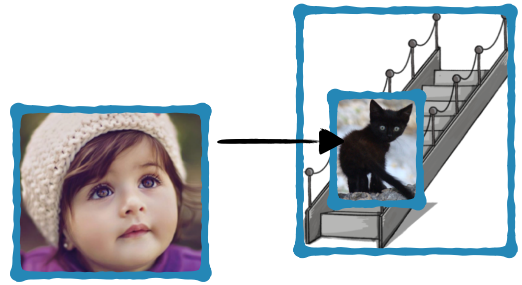
Intermediate representations

Event participant roles
=
Thematic roles

Agent, Experiencer, Patient, Theme, Goal, Source, Location…

Some linking theory proposals: Recap

UTAH has a mapping to three fixed thematic categories, which then map to syntactic positions.



The little girl *blicked* the kitten on the stairs.

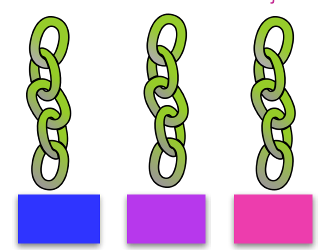
Syntactic positions

Intermediate representations

UTAH

fixed

Subject Object Oblique Object

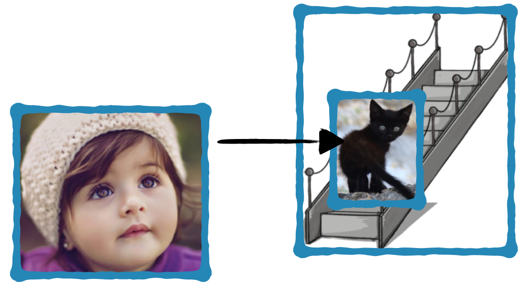


Event participant roles
=
Thematic roles

Agent, Experiencer, Patient, Theme, Goal, Source, Location…

Some linking theory proposals: Recap

rUTAH has a relative ordering of thematic roles, and maps those in relative order to syntactic positions.



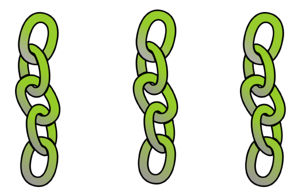
The little girl *blicked* the kitten on the stairs.

Syntactic positions

Intermediate representations

relative

Subject Object Oblique Object



First Second Third Highest

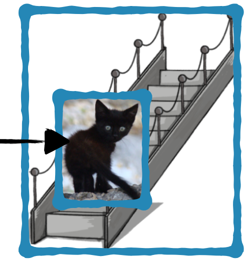
Agent > Experiencer >
Theme > Patient >
(Source, Goal, Location)

Event participant roles
=
Thematic roles

Agent, Experiencer, Patient, Theme, Goal, Source, Location…

Linking theory acquisition

Pearl & Sprouse 2021: the relative linking theory of rUTAH is easier to learn from English children's input.



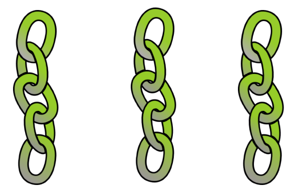
The little girl *blicked* the kitten on the stairs.

Syntactic positions

Intermediate representations

relative

Subject Object Oblique Object



First Second Third Highest

Agent > Experiencer >
Theme > Patient >
(Source, Goal, Location)

Event participant roles
=
Thematic roles

Agent, Experiencer, Patient, Theme, Goal, Source, Location…

Recap

Children rely on different strategies that often are helpful for figuring out a sentence's meaning, even before children know the syntactic rules of their language (world knowledge, order of mention).

However, sometimes these strategies don't work (ex: passives). Children have to learn the syntactic rules for how verbs combine with their arguments (like subjects and objects) in different configurations, like the passive. This takes time, and is affected by the verb's meaning.

More generally, children need to learn how a verb links its argument to the event participants — this is what linking theories do. Current evidence suggests children may find it easier to learn a relative linking theory like rUTAH, compared to one with fixed linking categories, like UTAH.

Questions?

You should be able to do up through question 15 on the review questions for syntactic acquisition, and up through question 8 on HW5.